

Topoi (ILF)

Goldblatt / Exercises / Reading group

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Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n , consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \dots \leq n-1$. How many arrows does $\mathbf{n}$ have?

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathcal{C}$ are valid subcategories?

- The empty category $\mathcal{C} = \emptyset$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. Verify that the collection $\mathcal{C}(X, X)$ of all $\mathcal{C}$ -arrows from X to X forms a monoid with respect to composition.

1.2.2 Sum Category

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The sum category, denoted $\mathcal{C} + \mathcal{D}$, is a category whose collection of objects is the disjoint union of $\mathcal{C}$ -objects and $\mathcal{D}$ -objects, and whose collection of arrows is the disjoint union of $\mathcal{C}$ -arrows and $\mathcal{D}$ -arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathcal{C}$ and $\mathcal{D}$. Verify that $\mathcal{C} + \mathcal{D}$ is a category.
- Verify that there are no $(\mathcal{C} + \mathcal{D})$ -arrows between any object that came from $\mathcal{C}$ and any object

that came from $\mathcal{D}$.

- Find an example of $\mathcal{C}$ and $\mathcal{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

1.3 Extra Exercise

1.3.1 Quivers

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \text{dom}(f)$, and $t(f) = \text{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.